

CLINICAL GENOME REPORT

Genome identifier: id-1492
Isolate/laboratory identifier: M54
Year of isolation: 2020
Location: Vadodara, India

Source type: Human
Host tissue sampled: Urine
Genome report date: 2026-04-17
Genome quality: Acceptable

Section 1: Clinical microbiology

Summary: *Klebsiella pneumoniae* with intrinsic ampicillin resistance. Note presence of ESBL and CPE genetic determinants, advise correlating with phenotypic antibiotic susceptibility testing.

Organism: *Klebsiella pneumoniae*

Bacterial typing: subspecies characterisation and classification to assess isolate similarity
<https://pubmlst.org/multilocus-sequence-typing>

MLST (Multilocus sequence typing)
Sequence type (ST): 16

Profile:	gapA	infB	mdh	pgi	phoE	rpoB	tonB
	2	1	2	1	4	4	4

Antibiotic resistance determinants: Genetically determined resistance mechanisms have been identified. Genetic variants (homologs) are highlighted *. "None found" means no resistance mechanisms have been identified in the antimicrobial resistance databases that exist at the time of reporting. It is best practice to review these with the phenotypic antibiotic susceptibility profile. <https://github.com/kebgonomics/Kleborate>.

Drug class	Acquired genotypes
Penicillins (expected resistance due to blaSHV)	blaSHV-11 [^] , blaCTX-M-15, blaNDM-5;OXA-181
+ β -lactamase inhibitor	None found
Cephalosporins (3 rd gen)	blaCTX-M-15, blaNDM-5;OXA-181
+ β -lactamase inhibitor	None found
Carbapenems	blaNDM-5;OXA-181
Porin mutations (multiple drug classes)	OmpK36:p.136_137insThrAsp;OmpK35:p.Ile11fs
Aminoglycosides	aac(3)-IIId [^] ;aadA [*] ;aadA2 [^] ;rmtB
Fluoroquinolones	qnrS1, GyrA:p.Ser83Phe;GyrA:p.Asp87Asn;ParC:p.Glu84Lys
Fosfomycin	None found
Phenicol	cmlA5
Polymixins	None found
Tigecycline	None found
Trimethoprim	dfrA12
+ Sulfonamides	sul1

Capsule and O typing: Polysaccharide K and lipopolysaccharide O serotypes as predicted by KL and O genotype <https://github.com/kebgonomics/Kaptive>.

K type: K81

O type: O13

Virulence factors: Factors that may lead to increased ability to cause invasive disease
<https://github.com/kebgonomics/Kleborate>.

Virulence score: 1

Acquired siderophores:

Aerobactin ST: None

Salmochelins ST: None

Yersiniabactin ST: 384-2LV

Other factors:

Colibactin ST: None

Hypermucoidy: None

Section 2: Public health and Infection prevention and control

Summary: *Klebsiella pneumoniae* not known to be related to other isolates in the same geography.

Bacterial typing: subspecies characterisation and classification to assess isolate similarity

<https://pubmlst.org/multilocus-sequence-typing>

MLST (Multilocus sequence typing)

Sequence type (ST): 16

Profile:

gapA	infB	mdh	pgi	phoE	rpoB	tonB
2	1	2	1	4	4	4

Phylogroup: Kp1

Sublineage: SL17

Clonal group: CG16

cgST: 12601

LIN (Life Identification Number): 0_0_22_27_0_7_0_0_0_0

Plasmid Inc typing: Plasmid replicons that are extracted from short sequence read data may be less reliable than from long read data <https://cge.food.dtu.dk/services/PlasmidFinder/>

Plasmid type: Col(pHAD28), Col440II, ColKP3, IncFIA(HI1), IncFII, IncR, IncX3

Outbreak analysis: Identifying isolates that are genetically similar and maybe part of a transmission chain

<https://bigsd.b.pasteur.fr/klebsiella/cgmlst-lincodes/>.

Allele based:

LIN (Life Identification Number): 0_0_22_27_0_7_0_0_0_0

(based on cgMLST scheme with 629 genes)

There are 4 related isolates identified with the same LINcode prefix [0_0_22_27_0](#) (representing the 10 core genome MLST distance threshold) in the database.

id	isolate	country	year	LINcode
22874	NMI4821/11	Poland	2011	0_0_22_27_0_0_5_0_0_0
22876	NMI10734/11	Poland	2011	0_0_22_27_0_0_5_0_0_0
22877	NMI10898/11	Poland	2011	0_0_22_27_0_0_5_0_0_0
1495	MyNCGM268	Myanmar	2016	0_0_22_27_0_6_0_0_0_0
1492	M54	India	2020	0_0_22_27_0_7_0_0_0_0

Most isolates with identical LIN codes (all 10 thresholds) have <100 and often <25 single nucleotide differences, but further single nucleotide analysis may be required for these isolates in cluster/outbreak investigations.